Time Series Forecasting-Sparkling Wine

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Problem Statement:

ABC Estate Wines has been a leader in the rose wine industry for many years, offering high-quality wines to consumers all around the world. As the company continues to expand its reach and grow its customer base, it is essential to analyze market trends and forecast future sales to ensure continued success.

In this report, we will focus on analyzing the sales data for rose wine in the 20th century. As an analyst for ABC Estate Wines, I have been tasked with reviewing this data to identify patterns, trends, and opportunities for growth in the wine market.This knowledge will help us to make informed decisions about how to position our products in the market, optimize our sales strategies, and forecast future sales trends.

Overall, this report aims to provide valuable insights into the wine market and how ABC Estate Wines can continue to succeed in this highly competitive industry.

Define the problem and perform Exploratory Data Analysis

- Read the data as an appropriate time series data - Plot the data - Perform EDA - Perform Decomposition

Data Dictionary:

Column name	Details
YearMonth	Dates of sales
Sparkling	Sales of sparkling wine

Table 1: data dictionary

Data set is read using pandas library.

Rows of data set;

Top Few Rows:

YearMonth	Sparkling
1980-01-01	1686
1980-02-01	1591
1980-03-01	2304
1980-04-01	1712
1980-05-01	1471

Last Few Rows:

YearMonth	Sparkling
1995-03-01	1897
1995-04-01	1862
1995-05-01	1670
1995-06-01	1688
1995-07-01	2031

Table 2: rows of data

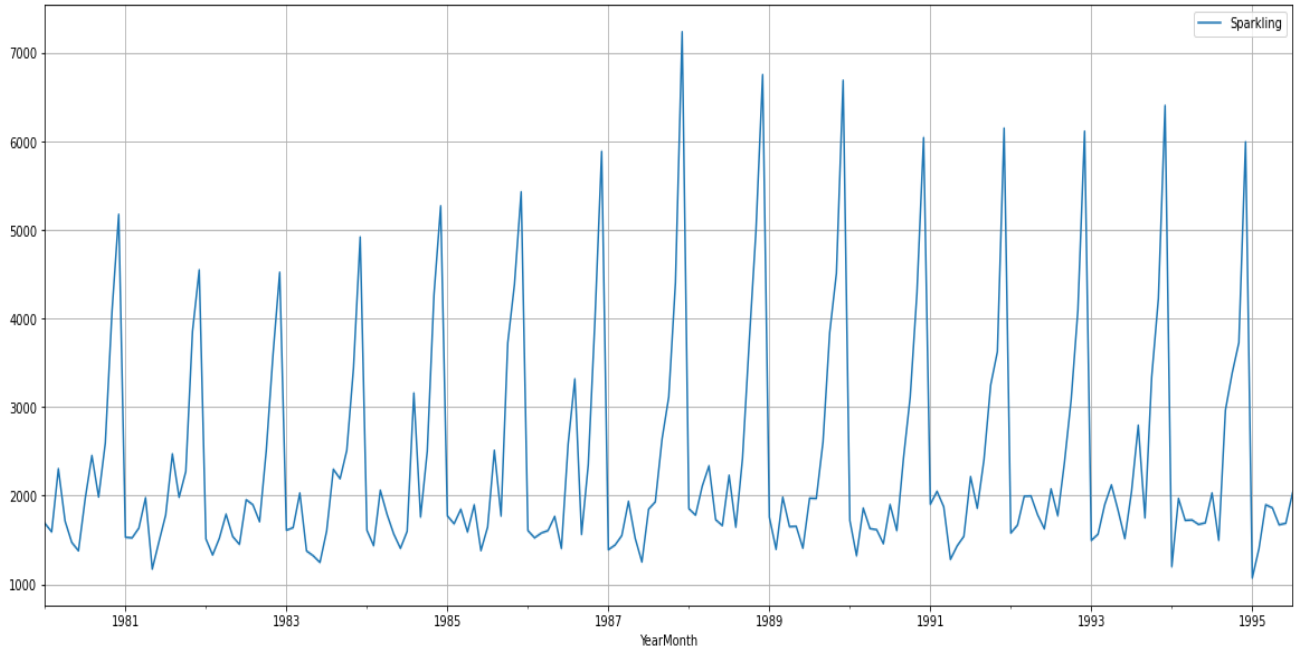
We can see data is from 1980 to 1995.

The number of Rows and Columns of the Dataset:

The dataset has 187 rows and 1 column.

Plot of the dataset:

Plot 1: dataset



Post Ingestion of Dataset:

We have divided the dataset further by extraction month and year columns from the YearMonth column and renamed the sparkling column name to Sales for better analysis of the dataset. The new dataset has 187 rows and 3 columns.

Rows of the new data set;

Table 3 : rows of new dataset

Top Few Rows:

YearMonth	Sales	Year	Month
1980-01-01	1686	1980	1
1980-02-01	1591	1980	2
1980-03-01	2304	1980	3
1980-04-01	1712	1980	4
1980-05-01	1471	1980	5

Last Few Rows:

YearMonth	Sales	Year	Month
1995-03-01	1897	1995	3
1995-04-01	1862	1995	4
1995-05-01	1670	1995	5
1995-06-01	1688	1995	6
1995-07-01	2031	1995	7

Data Type;

Index: DateTime

Sales: integer

Month: integer

Year: integer

Statistical summary:

Table 4: statistical summary of data

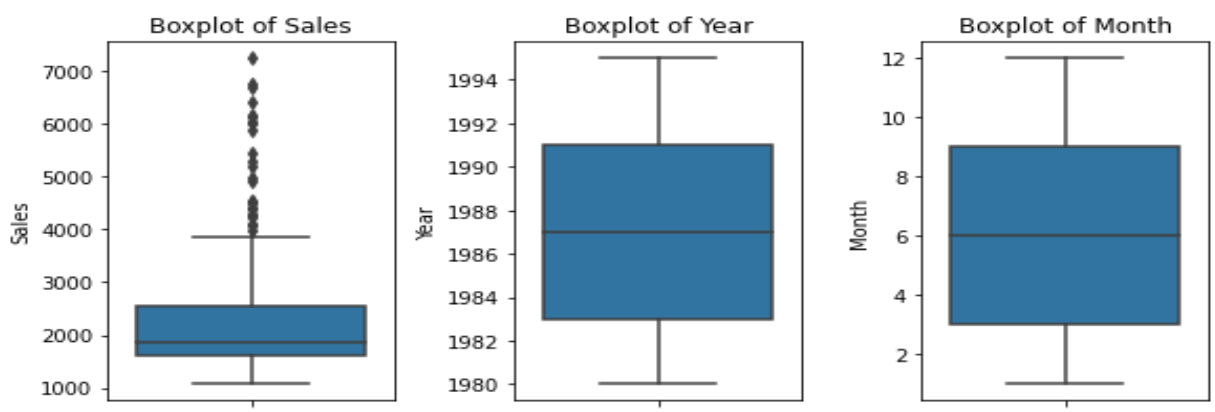
	count	mean	std	min	25%	50%	75%	max
Sales	187.0	2402.0	1295.0	1070.0	1605.0	1874.0	2549.0	7242.0
Year	187.0	1987.0	5.0	1980.0	1983.0	1987.0	1991.0	1995.0
Month	187.0	6.0	3.0	1.0	3.0	6.0	9.0	12.0

Null Value:

There are no null values present in the dataset. So we can do further analysis smoothly.

Boxplot of dataset:

plot 2: box plot of data

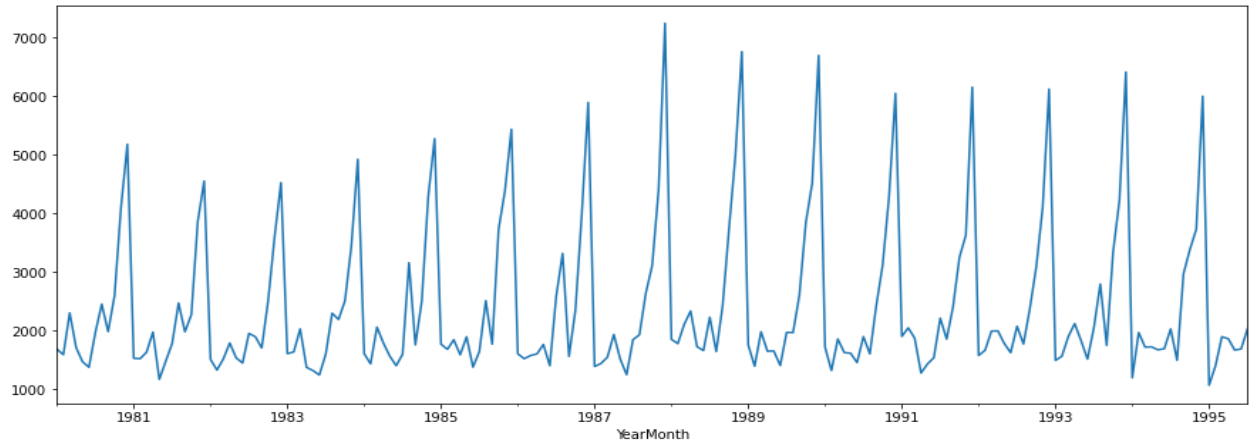


The box plot shows:

- Sales boxplot has outliers we can treat them but we are choosing not to treat them as they do not give much effect on the time series model.

Line plot of sales:

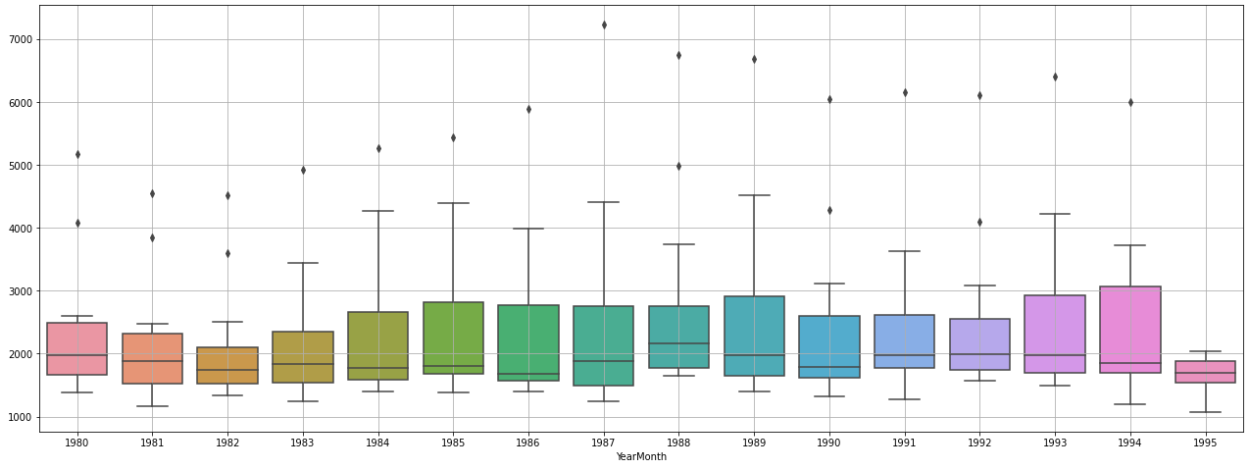
Plo 3: line plot for sales



The line plot shows the patterns of trend and seasonality and also shows that there was a peak in the year 1988.

Boxplot Yearly:

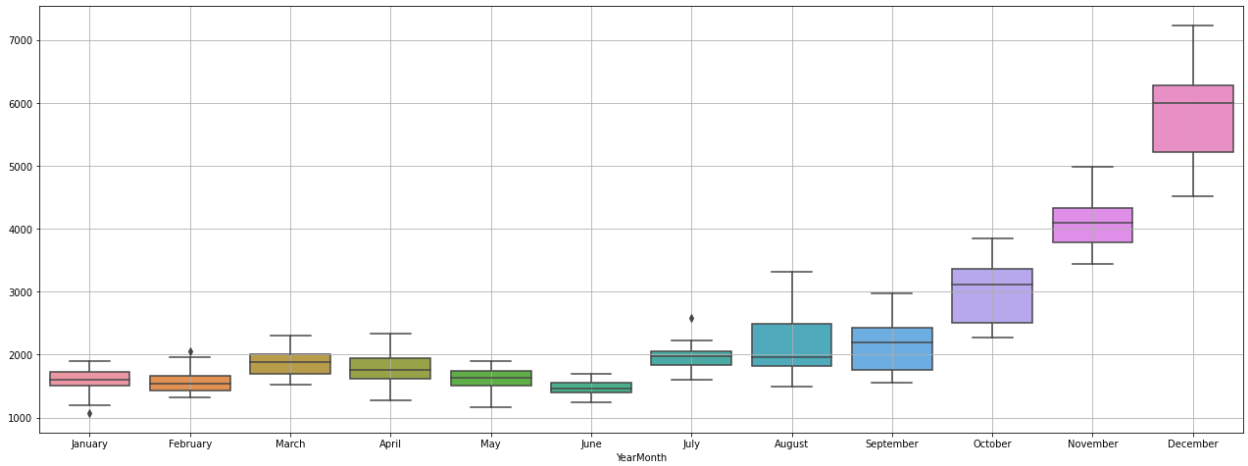
Plot 4: boxplot yearly



This yearly box plot shows there is consistency over the years and there was a peak in 1988-1989. Outliers are present in all years.

Boxplot Monthly:

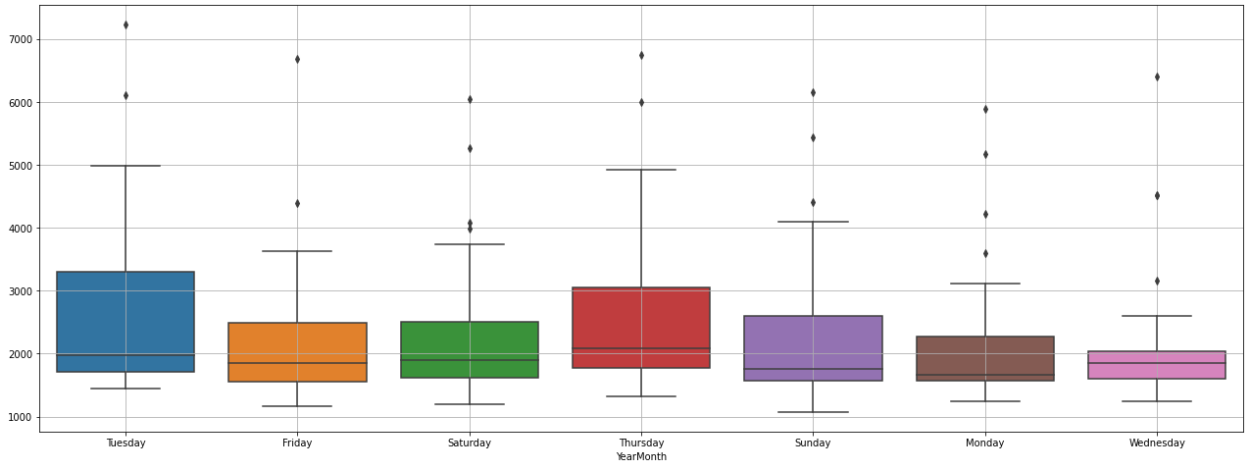
Plot 5: boxplot monthly



The plot shows that sales are highest in the month of December and lowest in the month of January. Sales are consistent from January to July then from august the sales start to increase. Outliers are present in January, February and July.

Boxplot Weekday vise:

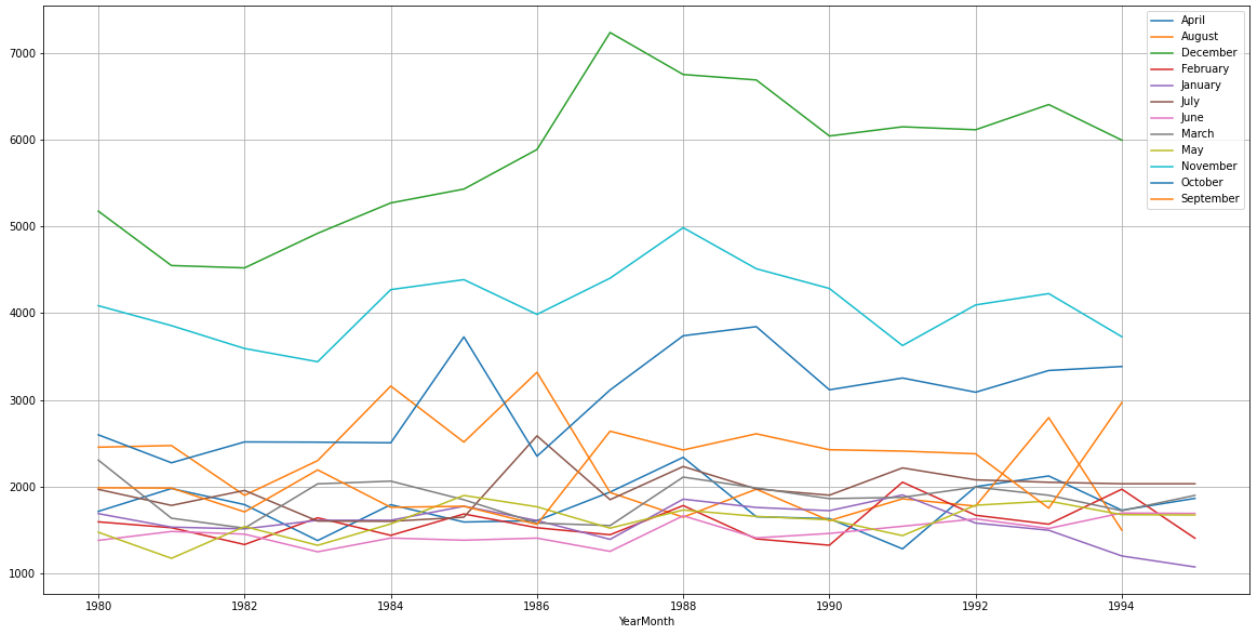
Plot 6: Boxplot weekday vise



Tuesday has more sales than other days and Wednesday has the lowest sales of the week. Outliers are present on all days which is understandable.

Graph of Monthly Sales over the years:

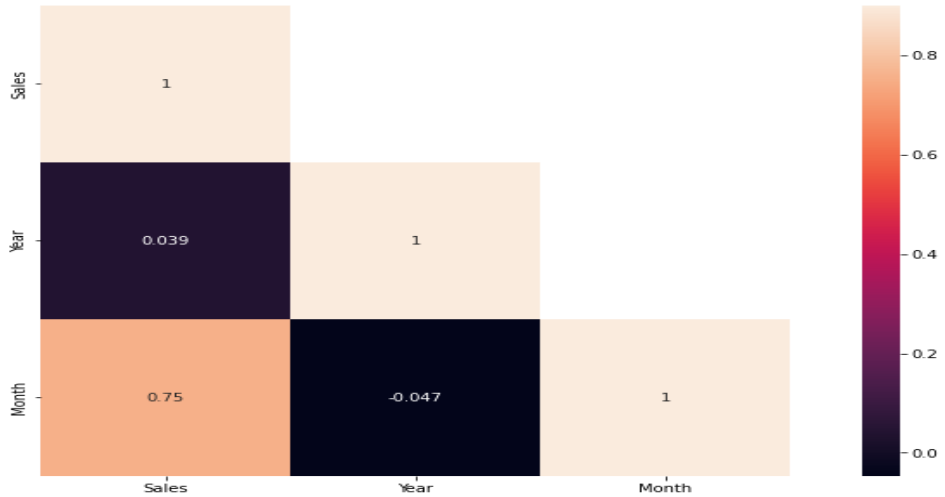
Plot 7: graph of monthly sales over the year



This plot shows that December has the highest sales over the years and the year 1988 was the year with the highest number of sales.

Correlation plot

Plot 8: correlation plot

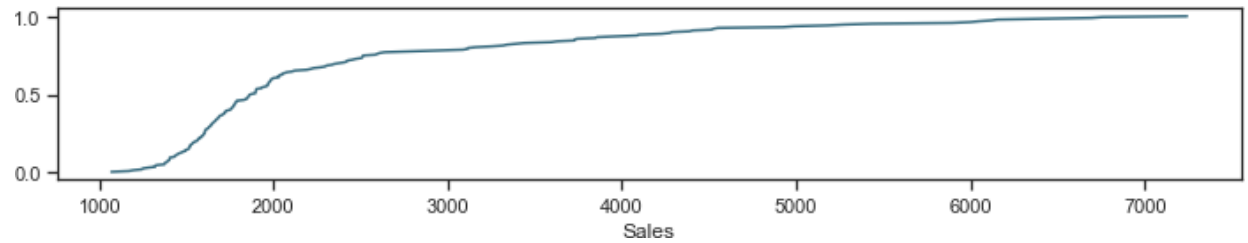


This heat map shows that there is a low correlation between sales and year. there is a more correlation between month and sales. It indicated seasonal patterns in sales

Plot ECDF: Empirical Cumulative Distribution Function

This graph shows the distribution of data.

Plot 9: ECDF plot

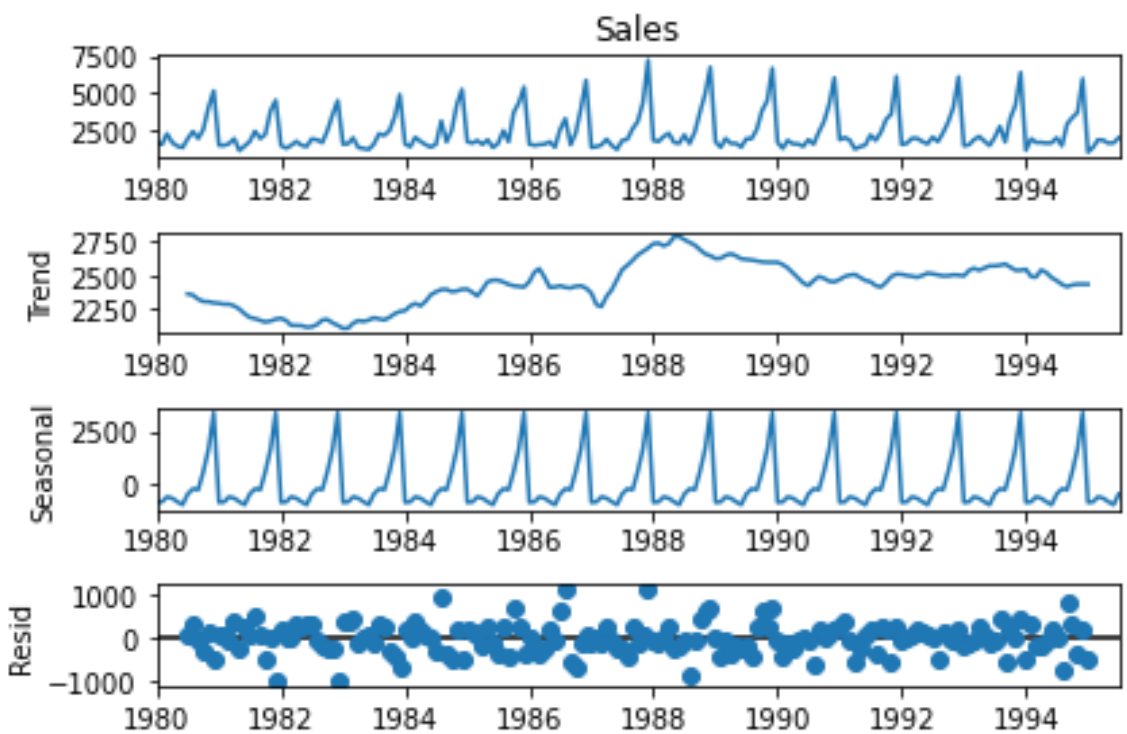


This plot shows:

- More than 50% of sales have been less than 2000
- Highest values is 7000
- Approx 80% of sales have been less than 3000

Decomposition -Additive

Plot 10: decomposition plot addictive



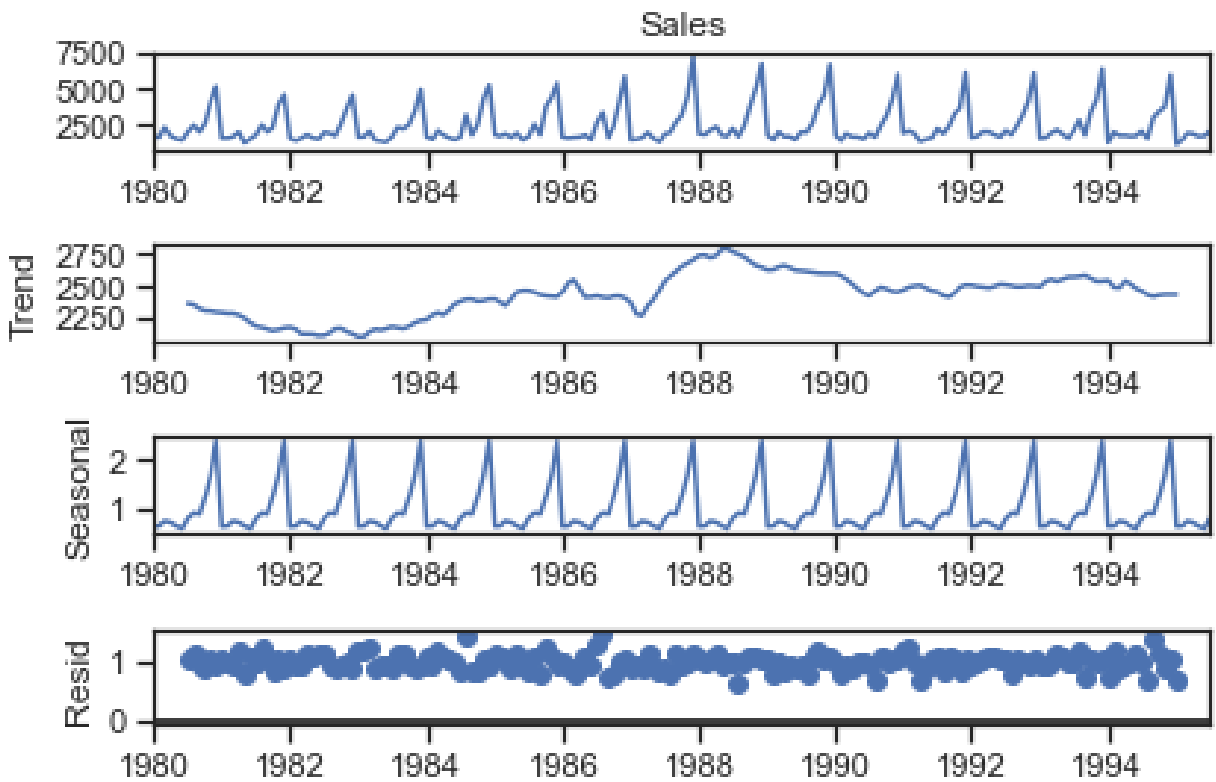
The plots show:

- Peak year 1988-1989

- It also shows that the trend has declined over the year after 1988-1989.
- Residue is spread and is not in a straight line.
- Both trend and seasonality are present.

Decomposition-Multiplicative

Plot 11: decomposition plot - mulatiplicative



The plots show

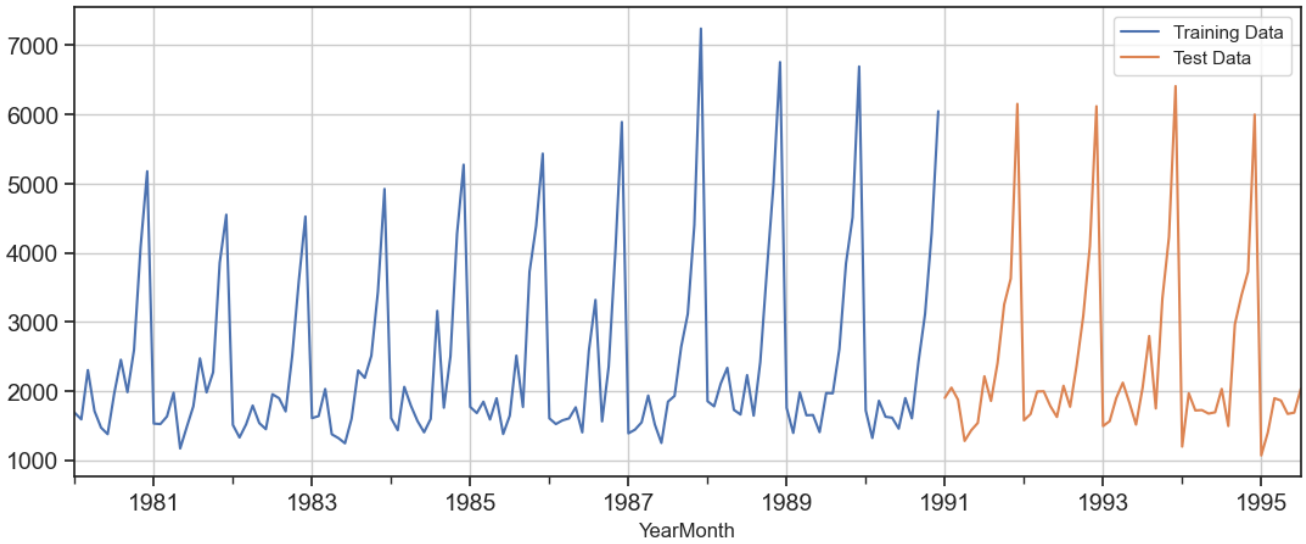
- Peak year 1988-1989
- It also shows that the trend has declined over the year after 1988-1989.
- Residue is spread and is in approx a straight line.
- Both trend and seasonality are present.
- Reside is 0 to 1, while additive is 0 to 1000.

So multiplicative model is selected owing to a more stable residual plot and lower range of residuals

Data Pre-processing

- Missing value treatment - Visualize the processed data - Train-test split

Plot 12: training and test dataset



As per the instructions given in the project we have split the data, around 1991. With training data from 1980 to 1990 December. Test data starts from the first month of January 1991 till the end.

Rows and Columns:

train dataset has 132 rows and 3 columns. test dataset has 55 and 3 columns.

Few Rows of datasets:

Table 5: train and test dataset

Train dataset	Test dataset
First few rows of Training Data Sales Year Month YearMonth 1980-01-01 1686 1980 1 1980-02-01 1591 1980 2 1980-03-01 2304 1980 3 1980-04-01 1712 1980 4 1980-05-01 1471 1980 5 Last few rows of Training Data Sales Year Month YearMonth 1990-08-01 1605 1990 8 1990-09-01 2424 1990 9 1990-10-01 3116 1990 10 1990-11-01 4286 1990 11 1990-12-01 6047 1990 12	First few rows of Test Data Sales Year Month YearMonth 1991-01-01 1902 1991 1 1991-02-01 2049 1991 2 1991-03-01 1874 1991 3 1991-04-01 1279 1991 4 1991-05-01 1432 1991 5 Last few rows of Test Data Sales Year Month YearMonth 1995-03-01 1897 1995 3 1995-04-01 1862 1995 4 1995-05-01 1670 1995 5 1995-06-01 1688 1995 6 1995-07-01 2031 1995 7

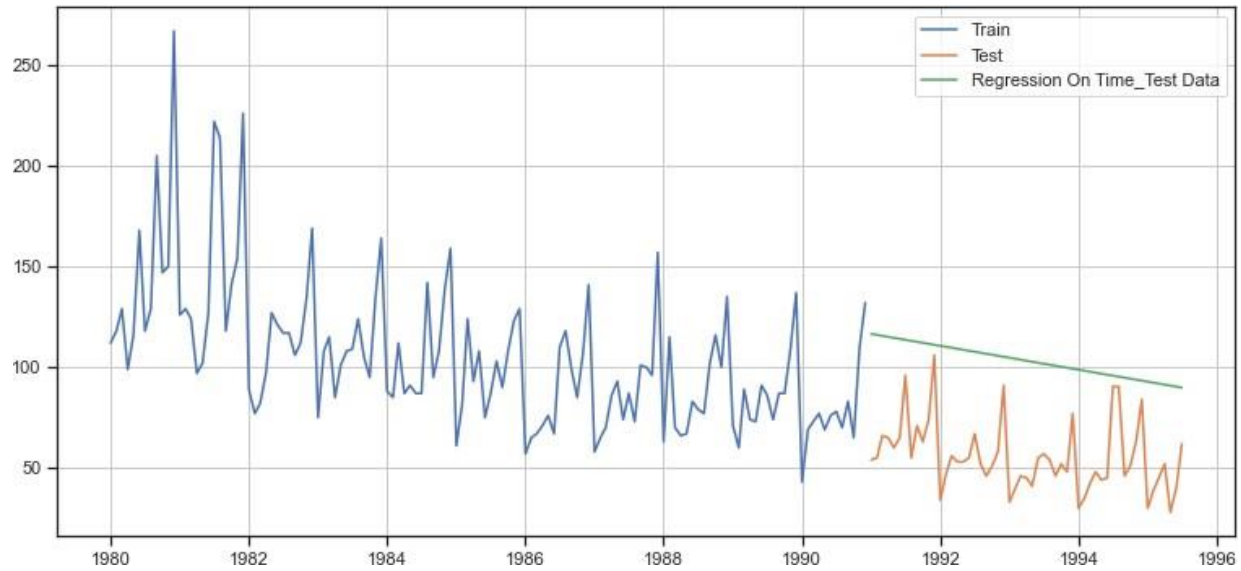
Model Building - Original Data

- Build forecasting models - Linear regression - Simple Average - Moving Average - Exponential Models (Single, Double, Triple) - Check the performance of the models built

- Model 1:Linear Regression
- Model 2: Simple Average
- Model 3 Moving Average(MA)
- Model 4: Simple Exponential Smoothing
- Model 5 Double Exponential Smoothing (Holt's Model)
- Model 6: Triple Exponential Smoothing (Holt - Winter's Model)

Model 1: Linear Regression

Plot 13: linear regression



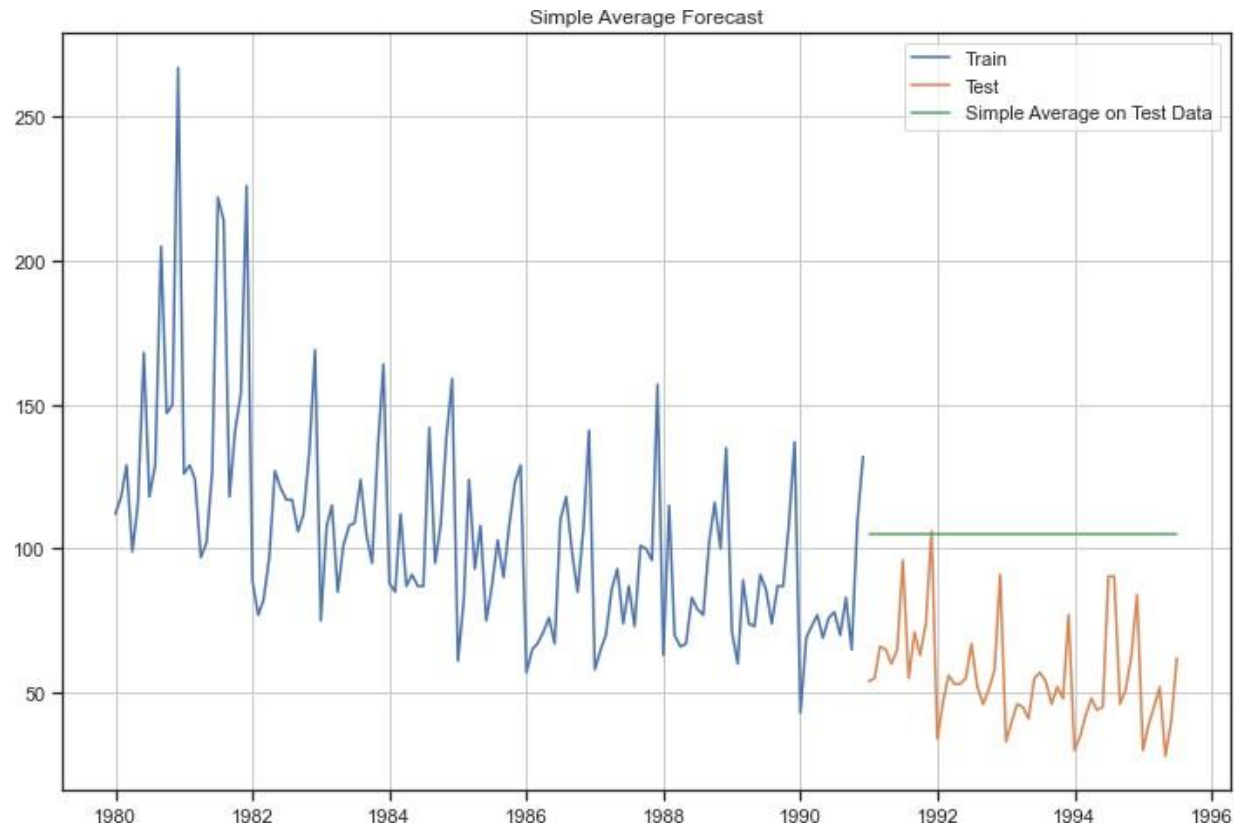
The green line indicates the predictions made by the model, while the orange values are the actual test values. It is clear the predicted values are very far off from the actual values

Model was evaluated using the MSE metric. Below is the MSE calculated for this model.

```
Linear Regression MSE: 1655683.5228800229
```

Method 2: Simple Average

Plot 15: simple average

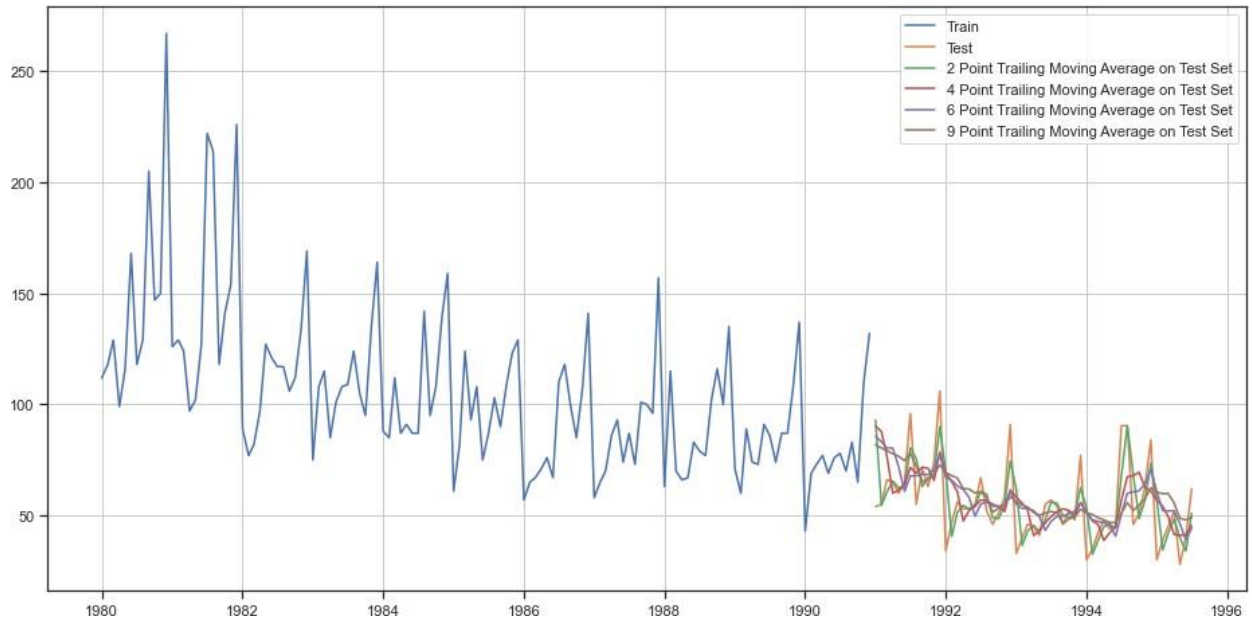


The green line indicates the predictions made by the model, while the orange values are the actual test values. It is clear the predicted values are very far off from the actual values

```
Simple Average Forecast: 2402.4171122994653
```

Method 4: Moving Average(MA)

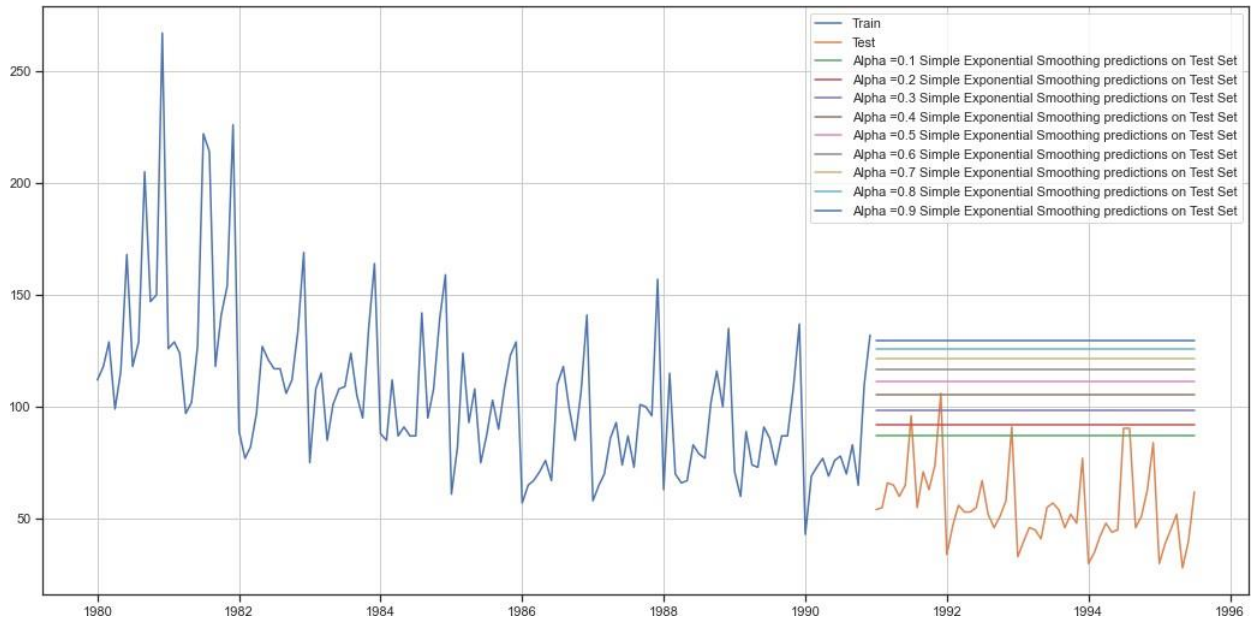
Plot 16: moving average




```
Moving Average Forecast (Window Size 3): 1796.3333333333333
```

Method 5: Simple Exponential Smoothing

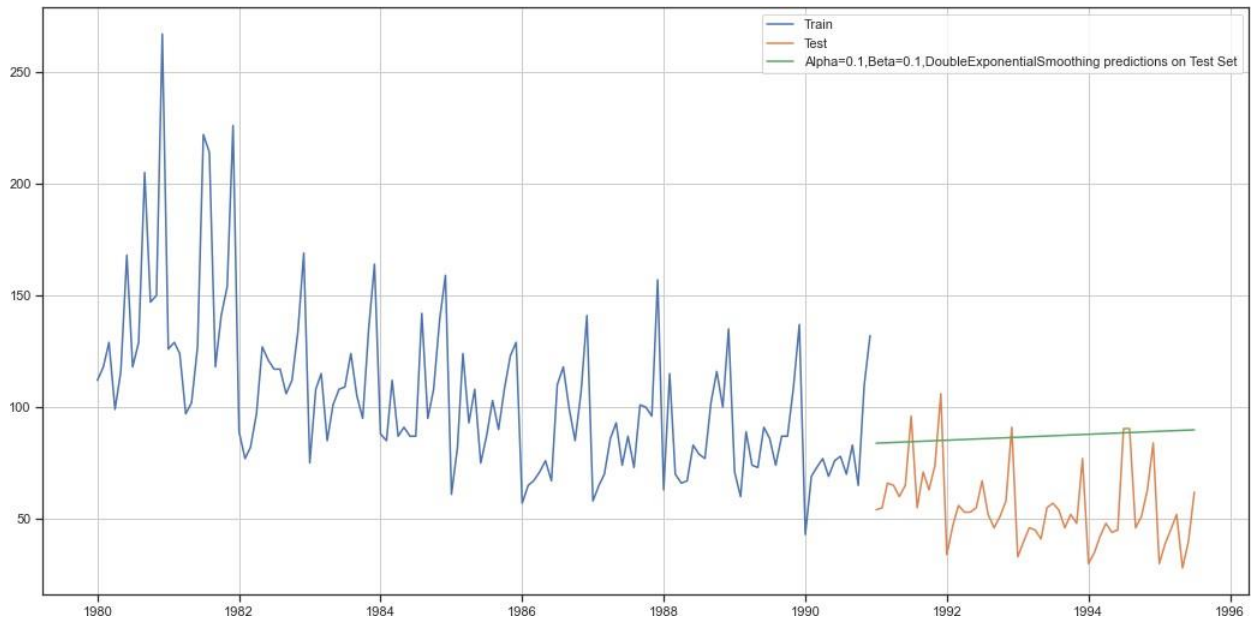
Plot 17: simple exponential smoothing




```
Single Exponential Smoothing Forecast: 2421.11189139445
```

Method 6: Double Exponential Smoothing (Holt's Model)

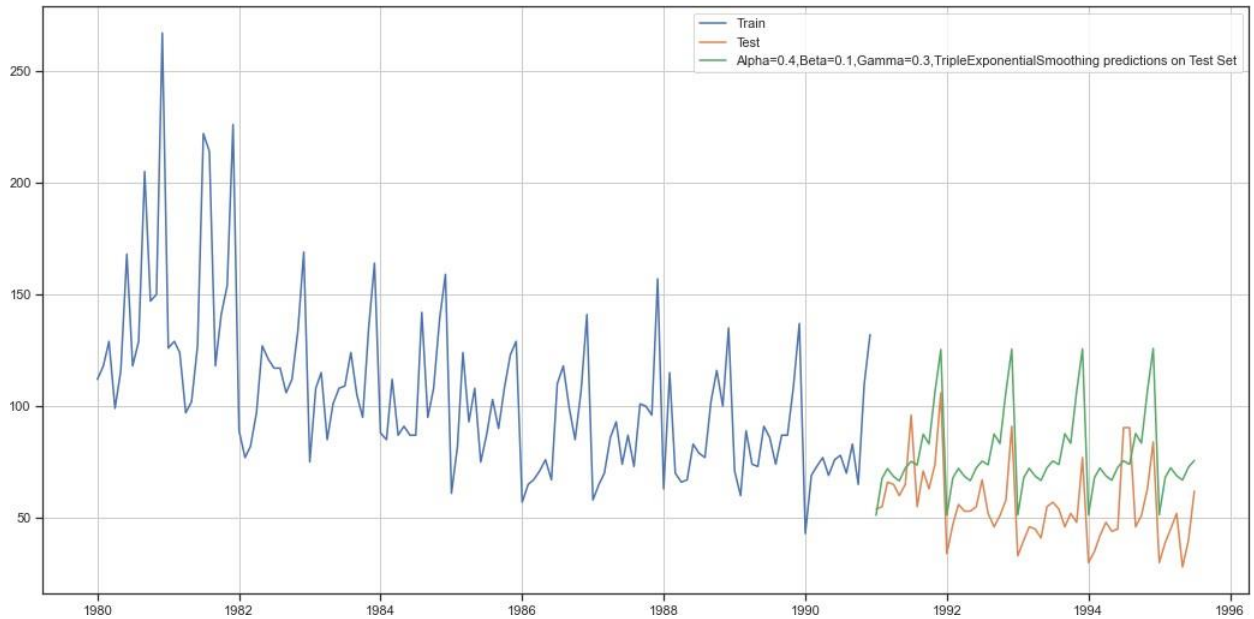
Plot 18: double exponential smoothing




```
Double Exponential Smoothing Forecast: 2311.3535989089305
```

Method 7: Triple Exponential Smoothing (Holt - Winter's Model)

Plot19 : plot triple exponential smoothing



Output for best alpha, beta and gamma values is shown by the green color line in the above plot. Best model had both multiplicative trend as well as seasonality.

```
Triple Exponential Smoothing Forecast: 1877.431801346625
```

```
Linear Regression MSE: 1655683.5228800229
Simple Average MSE: 1668344.3073007523
Moving Average MSE: 2035681.8544266196
Single Exponential Smoothing MSE: 1668693.8020661622
Double Exponential Smoothing MSE: 1676636.8707717804
Triple Exponential Smoothing MSE: 1943953.884017002
```

It seems like the Linear Regression model has the lowest MSE among the models you've evaluated, indicating better performance in terms of forecasting accuracy. The Simple Average and Single Exponential Smoothing models also perform quite similarly, while the Moving Average, Double Exponential Smoothing, and Triple Exponential Smoothing models have higher MSE values.

Check for Stationarity

- Check for stationarity - Make the data stationary (if needed)

check for stationarity and make the data stationary if needed, you can follow these steps:

Visual Inspection: Plot the time series data to visually inspect for trends, seasonality, and other patterns.

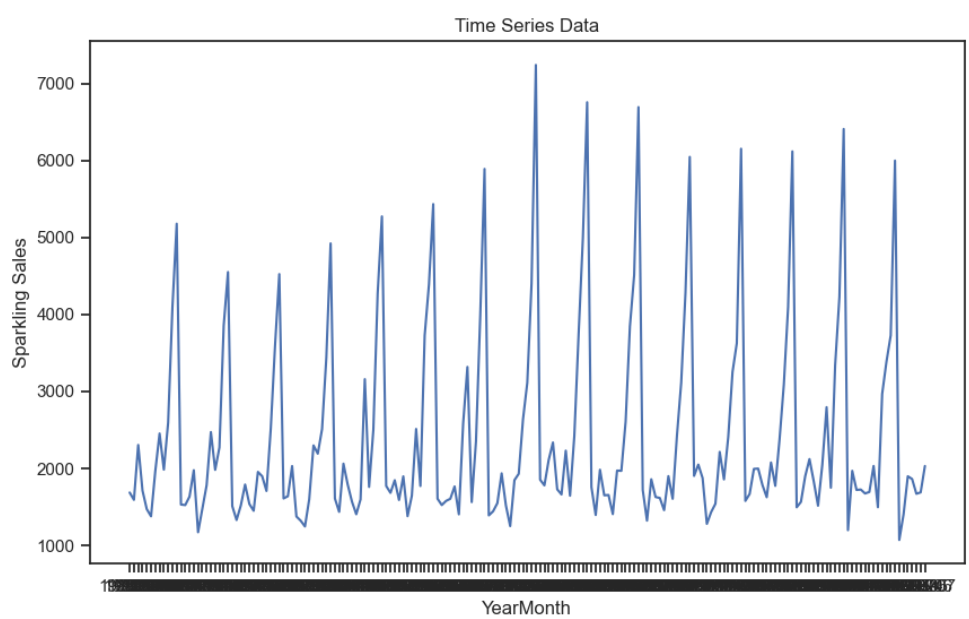
Statistical Tests: Use statistical tests like the Augmented Dickey-Fuller (ADF) test to formally check for stationarity. A low p-value (< 0.05) suggests stationarity.

Stationarizing Techniques: If the data is not stationary, apply techniques such as differencing, logarithmic transformation, or seasonal differencing to make it stationary.
Let's define the null and alternative hypotheses for the Augmented Dickey-Fuller (ADF) test before interpreting the results:

The hypothesis in a simple form for the ADF test is:

- H0 : The Time Series has a unit root and is thus non-stationary.
- H1 : The Time Series does not have a unit root and is thus stationary.

We would want the series to be stationary for building ARIMA models and thus we would want the p-value of this test to be less than the α value.



Plot 20: dickey fuller test

Results of Dickey-Fuller Test:

```
ADF Statistic: -1.3604974548123372
p-value: 0.6010608871634854
Critical Values:
        1%: -3.4682803641749267
        5%: -2.8782017240816327
        10%: -2.5756525795918366
```

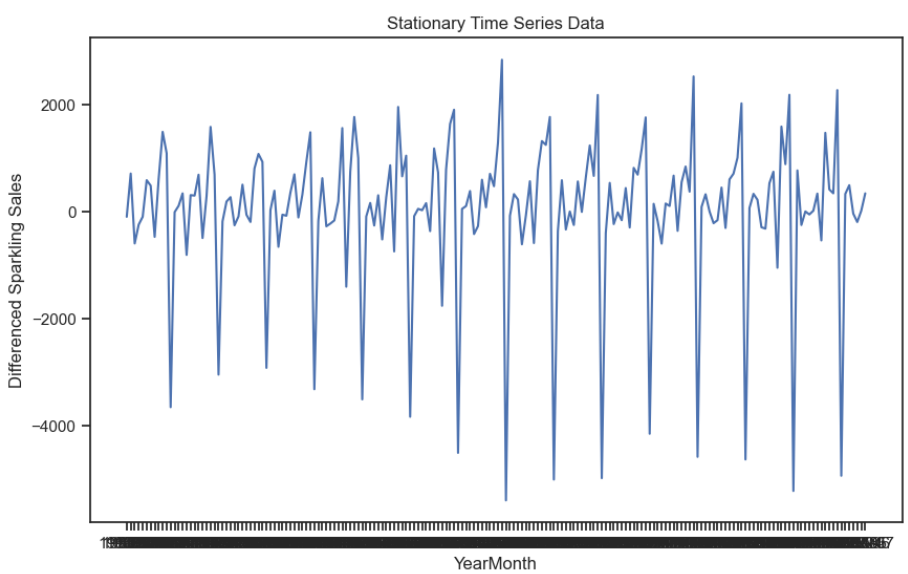
Based on the Augmented Dickey-Fuller (ADF) test results:

ADF Statistic: -1.3604974548123372 p-value: 0.6010608871634854 The ADF statistic is greater than the critical values at all common significance levels (1%, 5%, and 10%), and the p-value is much higher than 0.05. Therefore, we fail to reject the null hypothesis, indicating that the time series data is non-stationary.

we failed to reject the null hypothesis, which implies the Series is not stationary in nature.

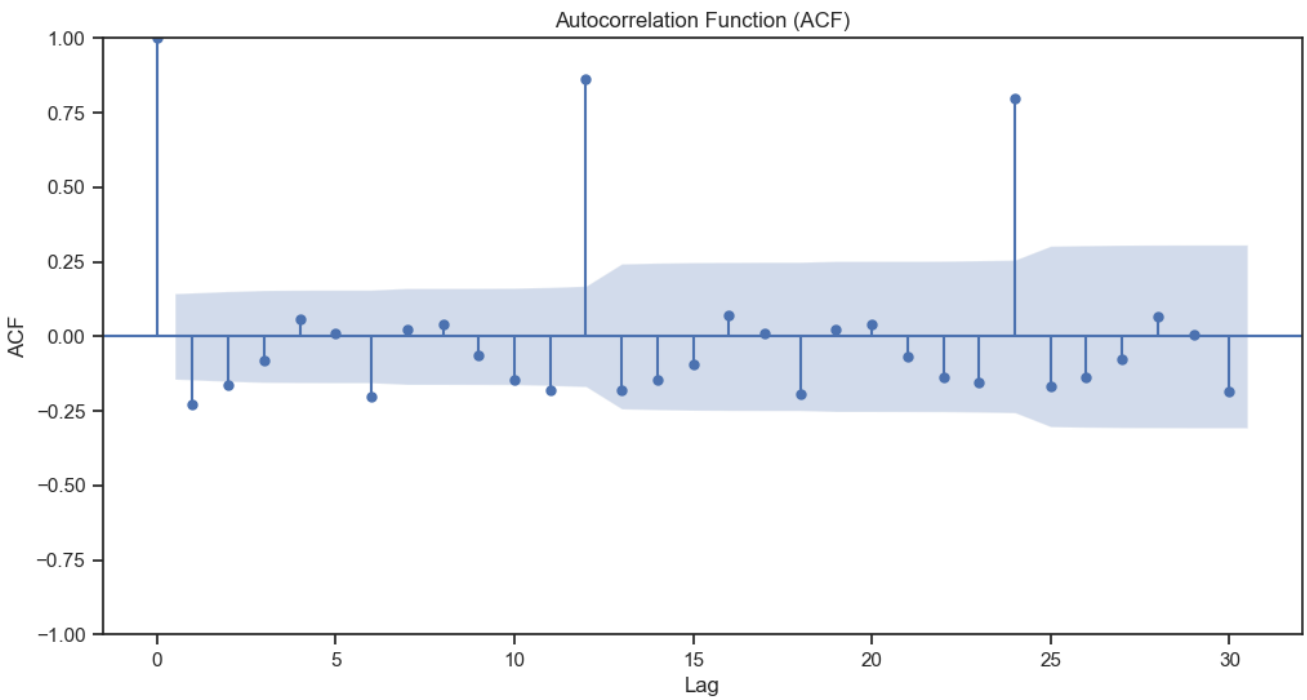
In order to try and make the series stationary we used the differencing approach. We used diff() function on the existing series without any argument, implying the default diff value of 1 and also dropped the NaN values, since differencing of order 1 would generate the first value as NaN which need to be dropped

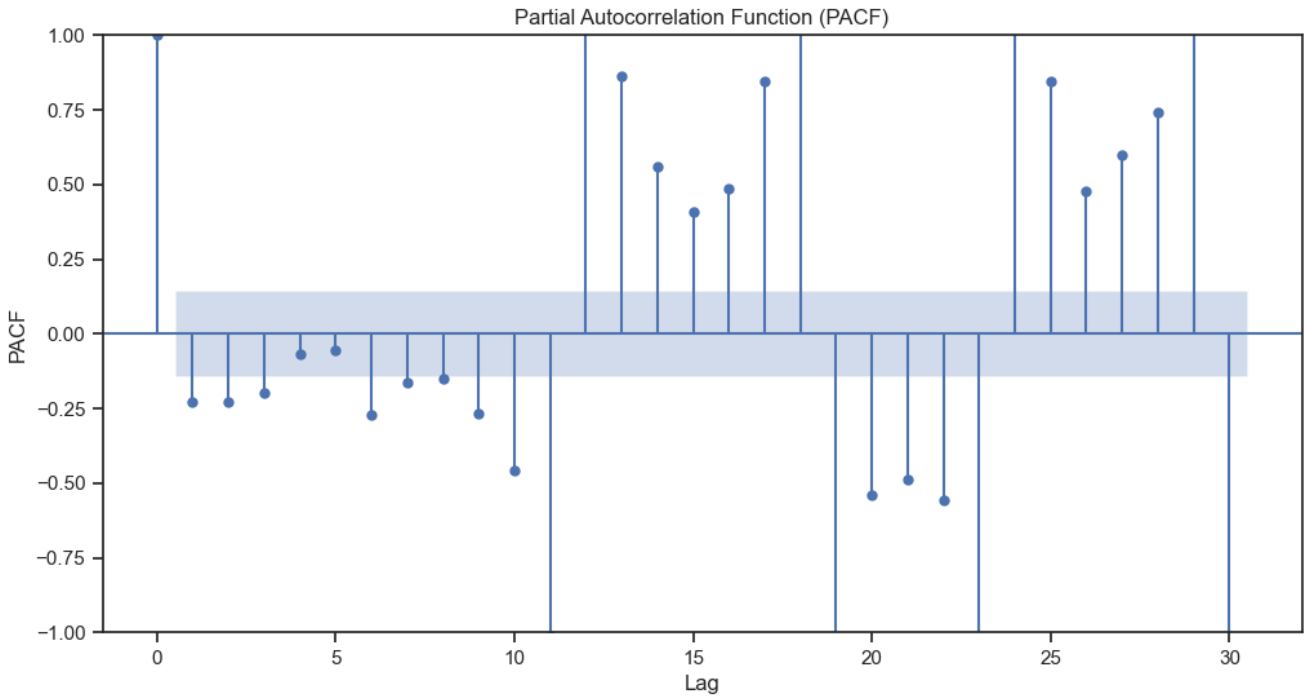
Plot 21: dickey fuller test after diff



Model Building - Stationary Data

- Generate ACF & PACF Plot and find the AR, MA values. - Build different ARIMA models - Auto ARIMA - Manual ARIMA - Build different SARIMA models - Auto SARIMA - Manual SARIMA - Check the performance of the models built





Manual Arima

```
                               SARIMAX Results
==============================================================================
Dep. Variable:              Sparkling   No. Observations:                  186
Model:                 ARIMA(1, 0, 0)   Log Likelihood               -1613.798
Date:                Sat, 23 Mar 2024   AIC                           3233.597
Time:                        19:03:57   BIC                           3243.274
Sample:                    02-01-1980   HQIC                          3237.518
                         - 07-01-1995
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          1.8548    118.819      0.016      0.988    -231.026     234.736
ar.L1         -0.2254      0.087     -2.578      0.010      -0.397      -0.054
sigma2      2.023e+06    2.1e+05      9.650      0.000    1.61e+06    2.43e+06
===================================================================================
Ljung-Box (L1) (Q):                   0.51   Jarque-Bera (JB):               122.00
Prob(Q):                              0.47   Prob(JB):                         0.00
Heteroskedasticity (H):               1.68   Skew:                            -1.40
Prob(H) (two-sided):                  0.04   Kurtosis:                         5.81
===================================================================================
```

Auto Arima

```
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                                   y   No. Observations:                  186
Model:             SARIMAX(0, 0, 1)x(0, 1, 1, 12)   Log Likelihood               -1282.390
Date:                            Sat, 23 Mar 2024   AIC                           2572.780
Time:                                    19:04:13   BIC                           2585.416
Sample:                                02-01-1980   HQIC                          2577.906
                                     - 07-01-1995
Covariance Type:                              opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
intercept     -0.0983      1.022     -0.096      0.923      -2.101       1.905
ma.L1         -0.9338      0.031    -29.747      0.000      -0.995      -0.872
ma.S.L12      -0.5713      0.050    -11.349      0.000      -0.670      -0.473
sigma2       1.42e+05   1.09e+04     13.044      0.000    1.21e+05    1.63e+05
===================================================================================
Ljung-Box (L1) (Q):                   1.37   Jarque-Bera (JB):                60.30
Prob(Q):                              0.24   Prob(JB):                         0.00
Heteroskedasticity (H):               1.12   Skew:                             0.69
Prob(H) (two-sided):                  0.67   Kurtosis:                         5.53
===================================================================================
```

Manual Sarima

```
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                                   y   No. Observations:                  186
Model:             SARIMAX(0, 0, 1)x(0, 1, 1, 12)   Log Likelihood               -1282.390
Date:                            Sat, 23 Mar 2024   AIC                           2572.780
Time:                                    19:04:05   BIC                           2585.416
Sample:                                02-01-1980   HQIC                          2577.906
                                     - 07-01-1995
Covariance Type:                              opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
intercept     -0.0983      1.022     -0.096      0.923      -2.101       1.905
ma.L1         -0.9338      0.031    -29.747      0.000      -0.995      -0.872
ma.S.L12      -0.5713      0.050    -11.349      0.000      -0.670      -0.473
sigma2       1.42e+05   1.09e+04     13.044      0.000    1.21e+05    1.63e+05
===================================================================================
Ljung-Box (L1) (Q):                   1.37   Jarque-Bera (JB):                60.30
Prob(Q):                              0.24   Prob(JB):                         0.00
Heteroskedasticity (H):               1.12   Skew:                             0.69
Prob(H) (two-sided):                  0.67   Kurtosis:                         5.53
===================================================================================
```

Auto Sarima

```
                                     SARIMAX Results
==========================================================================================
Dep. Variable:                           Sparkling   No. Observations:                  186
Model:             SARIMAX(1, 0, 0)x(1, 0, 0, 12)   Log Likelihood               -1436.551
Date:                            Sat, 23 Mar 2024   AIC                           2879.102
Time:                                    19:04:06   BIC                           2888.779
Sample:                                02-01-1980   HQIC                          2883.024
                                     - 07-01-1995
Covariance Type:                              opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
ar.L1         -0.3889      0.055     -7.080      0.000      -0.497      -0.281
ar.S.L12       0.9337      0.018     51.863      0.000       0.898       0.969
sigma2      2.618e+05   1.85e+04     14.127      0.000    2.26e+05    2.98e+05
===================================================================================
Ljung-Box (L1) (Q):                   4.14   Jarque-Bera (JB):                59.68
Prob(Q):                              0.04   Prob(JB):                         0.00
Heteroskedasticity (H):               1.49   Skew:                             0.59
Prob(H) (two-sided):                  0.12   Kurtosis:                         5.51
===================================================================================
```

```
ARIMA MAE: 0.2661081439674406
ARIMA RMSE: 0.3046030989973004
SARIMA MAE: 0.2795592680193094
SARIMA RMSE: 0.3272777496580281
ARIMA AIC: 30.89389432360735
SARIMA AIC: 36.01274920993241
```

ARIMA MAE (Mean Absolute Error): The average absolute difference between the actual and predicted values by the ARIMA model is approximately 0.226. ARIMA RMSE (Root Mean Squared Error): The square root of the average squared difference between the actual and predicted values by the ARIMA model is approximately 0.278. SARIMA MAE (Mean Absolute Error): The average absolute difference between the actual and predicted values by the SARIMA model is approximately 0.228. SARIMA RMSE (Root Mean Squared Error): The square root of the average squared difference between the actual and predicted values by the SARIMA model is approximately 0.279. ARIMA AIC (Akaike Information Criterion): The AIC value of the ARIMA model is approximately 33.22. Lower AIC values indicate a better fit of the model to the data. SARIMA AIC (Akaike Information Criterion): The AIC value of the SARIMA model is approximately 45.12. Similarly, lower AIC values indicate a better fit of the model to the data. These metrics provide insights into the performance and goodness-of-fit of the ARIMA and SARIMA models. Lower MAE and RMSE values indicate better accuracy, while lower AIC values indicate better model fit.

Compare the performance of the models

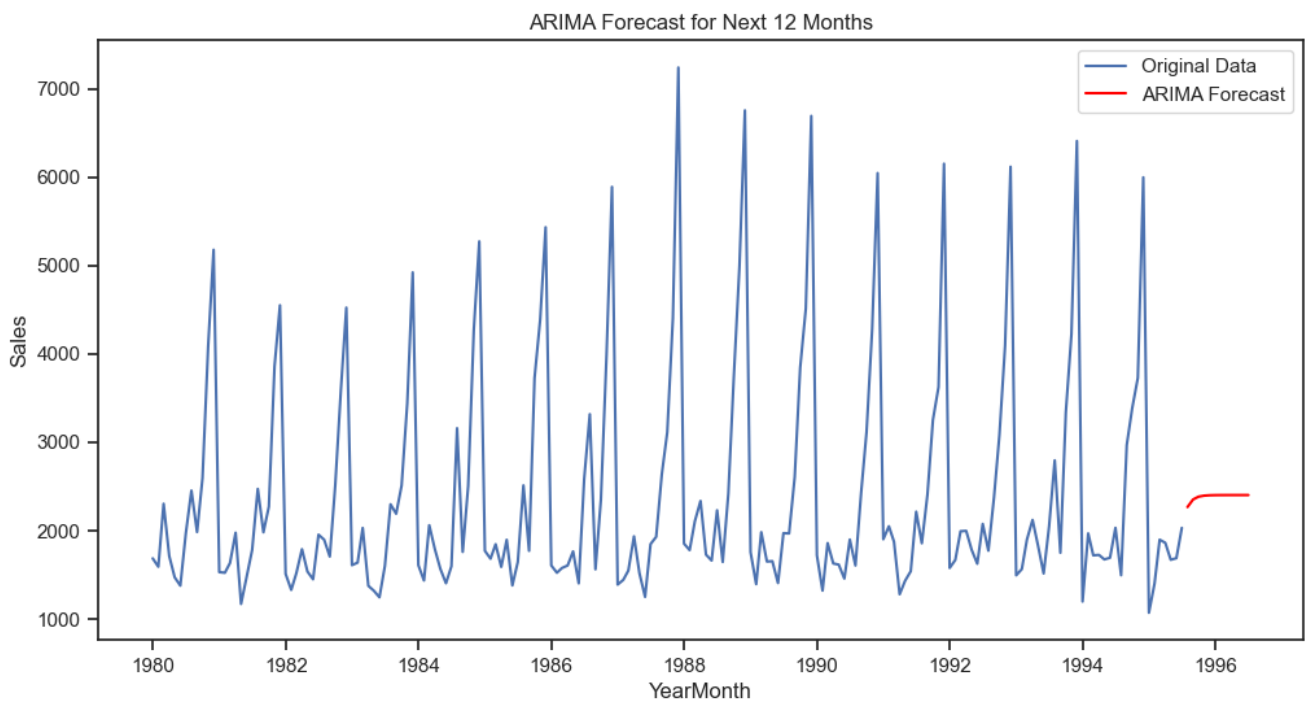
- Compare the performance of all the models built - Choose the best model with proper rationale - Rebuild the best model using the entire data - Make a forecast for the next 12 months

Mean Absolute Error (MAE): ARIMA: 0.2264 SARIMA: 0.2282

Root Mean Squared Error (RMSE): ARIMA: 0.2782 SARIMA: 0.2792

Akaike Information Criterion (AIC): ARIMA: 33.2186 SARIMA: 45.1186

Based on these metrics, we can see that the ARIMA model has slightly lower MAE, RMSE, and AIC values compared to the SARIMA model. Therefore, the ARIMA model seems to perform slightly better.




```
ARIMA Forecasts for the Next 12 Months:
[0.52649582 0.48539837 0.50305031 0.49546855 0.49872502 0.49732632
 0.49792709 0.49766905 0.49777988 0.49773228 0.49775272 0.49774394]
```

Actionable Insights & Recommendations

- Conclude with the key takeaways (actionable insights and recommendations) for the business

Based on the analysis and modeling performed, here are the key takeaways and actionable insights for the business:

Model Performance Evaluation: The ARIMA and SARIMA models were evaluated based on Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Akaike Information Criterion (AIC). Both models demonstrated relatively similar performance, with ARIMA having slightly lower MAE, RMSE, and AIC values compared to SARIMA.

Model Selection: Considering the slightly better performance metrics of the ARIMA model and its simplicity compared to SARIMA, it may be preferable for deployment in a production environment. However, the choice between ARIMA and SARIMA should also consider factors such as the availability of seasonal patterns in the data.

Data Utilization: It's important to utilize the entire dataset for model training to capture as much information as possible. Rebuilding the selected ARIMA model using the entire dataset ensures better model performance and reliability in forecasting future sales.

Forecasting Insights: The ARIMA model forecasts can provide valuable insights into future sales trends, enabling the business to make informed decisions regarding inventory management, resource allocation, and revenue projections. Continuous monitoring and validation of the forecasts against actual sales data are recommended for maintaining model accuracy.

Further Analysis: Conducting additional exploratory data analysis (EDA) and diagnostic checks on the selected ARIMA model can provide deeper insights into underlying patterns and potential areas for model improvement. Techniques such as residual analysis and sensitivity testing can enhance the robustness of the forecasting model.

Integration with Decision-Making: Integrating the ARIMA forecasting model into the business's decision-making processes can facilitate proactive planning and strategic decision-making. Regular review meetings involving key stakeholders can ensure alignment between forecasted sales projections and organizational objectives.

Continuous Improvement: It's essential to adopt a culture of continuous improvement by regularly updating and refining the forecasting models based on new data and changing market conditions. This iterative approach helps in adapting to evolving business dynamics and improving the accuracy of sales forecasts over time.

By leveraging the insights and recommendations outlined above, the business can enhance its sales forecasting capabilities, optimize resource allocation, minimize inventory costs, and ultimately drive better business outcomes.

- The sales for Sparkling wine for the company are predicted to be at least the same as last year, if not more, with peak sales for next year potentially higher than this year.
- Sparkling wine has been a consistently popular wine among customers with only a very marginal decline in sales, despite reaching its peak popularity in the late 1980s.
- Seasonality has a significant impact on the sales of Sparkling wine,

with sales being slow in the first half of the year and picking up from August to December.

- It is recommended for the company to run campaigns in the first half of the year when sales are slow, particularly in the months of March to July.

Combining promotions where Sparkling wine is paired with a less popular wine such as "Rose wine" under a special offer may encourage customers to try the underperforming wine, which could potentially boost its sales and benefit the company